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A surveillance vendor was caught exploiting a new SS7 attack to track people's phone locations

Security researchers say they have caught a surveillance company in the Middle East exploiting a new attack capable of tricking phone operators into disclosing a cell subscriber's location.

The attack relies on bypassing security protections that carriers have put in place to protect intruders from accessing SS7, or Signaling System 7, a private set of protocols used by the global phone carriers to route subscribers' calls and text messages around the world.

SS7 also allows the carriers to request information about which cell tower a subscriber's phone is connected to, typically used for accurately billing customers when they call or text someone from overseas, for example.

Researchers at Enea, a cybersecurity company that provides protections for phone carriers, said [this week](#) that they have observed the unnamed surveillance vendor exploiting the new bypass attack as far back as late 2024 to obtain the locations of people's phones without their knowledge.

Enea VP of Technology Cathal Mc Daid, who co-authored the blog post, told TechCrunch that the company observed the surveillance vendor target "just a few subscribers" and that the attack did not work against all phone carriers.

Mc Daid said that the bypass attack allows the surveillance vendor to locate an individual to the nearest cell tower, which in urban or densely populated areas could be narrowed to a few hundred meters.

Enea notified the phone operator it observed the exploit being used in, but declined to name the surveillance vendor, except to note it was based in the Middle East.

Mc Daid told TechCrunch that the attack was part of an increasing trend in malicious operators using these kinds

of exploits to obtain a person’s location, warning that the vendors behind their use “would not be discovering and using them if they were not successful somewhere.”

“We anticipate that more will be found and used,” Mc Daid said.

Surveillance vendors, which can include [spyware makers](#) and providers of bulk internet traffic, are private companies that typically work exclusively for government customers to conduct intelligence-gathering operations against individuals. Governments often claim to [use spyware](#) and other [exploitative technologies](#) against serious criminals, but the tools have also been used to target members of civil society, including [journalists](#) and [activists](#).

In the past, surveillance vendors have gained access to SS7 by way of a local phone operator, a misused [leased “global title,”](#) or through a government connection.

But due to the nature of these attacks happening at the cell network level, there is little that phone subscribers can do to defend against exploitation. Rather, defending against these attacks rests largely on the telecom companies.

In recent years, phone companies have installed firewalls and other cybersecurity protections to defend against SS7 attacks, but the patchwork nature of the global cell network means that not all carriers are as protected as others, including in the United States.

According to [a letter](#) sent to Sen. Ron Wyden’s office last year, the U.S. Department of Homeland Security said as far back as 2017 that several countries, notably China, Iran, Israel, and Russia, have used vulnerabilities in SS7 to “exploit U.S. subscribers.” Saudi Arabia has also been [found abusing flaws](#) in SS7 to conduct surveillance of its citizens in the United States.

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Saudi spies tracked phones using flaws the FCC failed to fix for years

Zack Whittaker

3:38 PM PDT · March 29, 2020



Lawmakers and security experts have long warned of security flaws in the underbelly of the world's cell networks. Now a whistleblower says the Saudi government is exploiting those flaws to track its citizens across the U.S. as part of a "systematic" surveillance campaign.

It's the latest tactic by the Saudi kingdom to spy on its citizens overseas. The kingdom has faced accusations of using [powerful mobile spyware](#) to hack into the phones of dissidents and activists to monitor their activities, including those close to Jamal Khashoggi, the Washington Post columnist who was murdered by agents of the Saudi regime. The kingdom also allegedly [planted spies at Twitter](#) to surveil critics of the regime.

The Guardian obtained [a cache of data](#) amounting to millions of locations on Saudi citizens over a four-month period beginning in November. The report says the location tracking requests were made by Saudi's three largest cell carriers — believed to be at the behest of the Saudi government — by exploiting weaknesses in SS7.

SS7, or Signaling System 7, is a set of protocols — akin to a private network used by carriers around the world — to route and direct calls and messages between networks. It's the reason why a T-Mobile customer can call an AT&T phone, or text a friend on Verizon — even when they're in another country. But experts say that weaknesses in the system have allowed attackers with access to the carriers — almost always governments or the carriers themselves — to listen in to calls and read text messages. SS7 also allows carriers to track the location of devices to just a few hundred feet in densely populated cities by making a "provide subscriber information" (PSI) request. These PSI requests are typically to ensure that the cell user is being billed correctly, such as if they are roaming on a carrier in another country.

Requests made in bulk and excess can indicate location tracking surveillance.

But despite years of warnings and numerous reports of attacks exploiting the system, the largest U.S. carriers have done little to ensure that foreign spies cannot abuse their networks for surveillance.

One Democratic lawmaker puts the blame squarely in the Federal Communication Commission's court for failing to compel cell carriers to act.

“I’ve been raising the alarm about security flaws in U.S. phone networks for years, but FCC chairman Ajit Pai has made it clear he doesn’t want to regulate the carriers or force them to secure their networks from foreign government hackers,” said Sen. Ron Wyden, a member of the Senate Intelligence Committee, in a statement on Sunday. “Because of his inaction, if this report is true, an authoritarian government may be reaching into American wireless networks to track people inside our country,” he said.

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A spokesperson for the FCC, the agency responsible for regulating the cell networks, did not respond to a request for comment.

A long history of feet-dragging

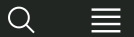
Wyden is not the only lawmaker to express concern. In 2016, Rep. Ted Lieu, then a freshman congressman, gave a security researcher permission to [hack his phone](#) by exploiting weaknesses in SS7 for an episode of CBS’ *60 Minutes*.

Lieu accused [the FCC](#) of being “guilty of remaining silent on wireless network security issues.”

The same vulnerabilities were [used a year later](#) in 2017 to drain the bank accounts of unsuspecting victims by intercepting and stealing the two-factor authentication codes necessary to log in sent by text message. The breach was one of the reasons why the U.S. government's standards and technology units, NIST, recommended [moving away](#) from using text messages to send two-factor codes.

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T-Mobile did not respond to a request for comment. Verizon (which owns TechCrunch) also did not comment. AT&T said at the time it “continually works with industry associations and government agencies” to address SS7 issues.

Fixing SS7

Fixing the problems with SS7 is not an overnight job. But without a regulator pushing for change, the carriers aren't inclined to budge.

Experts say [those same firewalls](#) put in place by the cell carriers can filter potentially malicious traffic and prevent some abuse. But an FCC working group tasked with understanding the risks posed by SS7 flaws in 2016 acknowledged that the vast majority of SS7 traffic is legitimate. “Carriers need to be measured as they implement solutions in order to avoid collateral network impacts,” [the report says](#).

In other words, it's not a feasible solution if it blocks real carrier requests.

Cell carriers have been less than forthcoming with their plans to fix their SS7 implementations. Only AT&T provided comment, telling The Guardian that it had “security controls to block location-tracking messages from roaming partners.” To what extent remains unclear, or if those measures will even help. Few experts have expressed faith in newer systems like Diameter, a

similar routing protocol for 4G and 5G, given there have already been [a raft](#) of [vulnerabilities](#) found in the newer system.

End-to-end encrypted apps, like Signal and WhatsApp, have made it harder for spies to snoop on calls and messages. But it's not a panacea. As long as SS7 remains a fixture underpinning the very core of every cell network, tracking location data will remain fair game.

[Privacy hawks in Congress call on Homeland Security to warn Americans of SS7 hacking threat](#)

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Google says surveillance vendor targeted Samsung phones with zero-days

Zack Whittaker — 1:00 PM PST · November 10, 2022

Google says it has evidence that a commercial surveillance vendor was exploiting three zero-day security vulnerabilities found in newer Samsung smartphones.

The vulnerabilities, discovered in Samsung's custom-built software, were used by attackers as part of an exploit chain to target Samsung phones running Android. The chained vulnerabilities allowed an attacker to gain kernel read and write privileges as the root user, and ultimately expose a device's data.

IMAGE CREDITS: VEEANNE CAO / TECHCRUNCH

Google Project Zero security researcher Maddie Stone said in a [blog post](#) that the exploit chain targets Samsung phones with a Exynos chip running a specific kernel version. Samsung phones are sold with Exynos chips primarily across Europe, the Middle East, and Africa, which is likely where the targets of the surveillance are located.

Stone said Samsung phones running the affected kernel at the time include [the S10](#), A50, and A51.

The flaws, since patched, were exploited by a malicious Android app, which the user may have been tricked into installing from outside of the app store. The malicious app allows the attacker to escape the app sandbox designed to contain its activity, and access the rest of the device's operating system. Only a component of the exploit app was obtained, Stone said, so it isn't known what the final payload was, even if the three vulnerabilities paved the way for its eventual delivery.

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“The first vulnerability in this chain, the [arbitrary file read and write](#), was the foundation of this chain, used four different times and used at least once in each step,” wrote Stone. “The Java components in Android devices don’t tend to be the most popular targets for security researchers despite it running at such a privileged level,” said Stone.

Google declined to name the commercial surveillance vendor, but said the exploitation follows a pattern similar to recent device infections where malicious Android apps were abused to deliver powerful nation-state spyware.

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Earlier this year security researchers discovered Hermit, an [Android and iOS spyware](#) developed by RCS Lab and used in targeted attacks by governments, with known victims in Italy and Kazakhstan. Hermit relies on tricking a

target into downloading and installing the malicious app, such as a disguised cell carrier assistance app, from outside of the app store, but then silently steals a victim's contacts, audio recordings, photos, videos, and granular location data. Google began notifying Android users whose [devices have been compromised by Hermit](#). Surveillance vendor Connexxa also used [malicious sideloaded apps](#) to target both Android and iPhone owners.

Google reported the three vulnerabilities to Samsung in late 2020, and Samsung rolled out patches to affected phones in March 2021, but did not disclose at the time that the vulnerabilities were being actively exploited. Stone said that Samsung has since committed to begin disclosing when vulnerabilities are actively exploited, following [Apple](#) and [Google](#), which also disclose in their security updates when vulnerabilities are under attack.

“The analysis of this exploit chain has provided us with new and important insights into how attackers are targeting Android devices,” Stone added, intimating that further research could unearth new vulnerabilities in custom software built by Android device makers, like Samsung.

“It highlights a need for more research into manufacturer specific components. It shows where we ought to do further variant analysis,” said Stone.

[Google is notifying Android users targeted by Hermit government-grade spyware](#)

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